

CHANUKA KEERTHISINGHA

Full Stack Software Engineer | Business Systems, AI & Automation

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PROFESSIONAL SUMMARY

Full Stack Software Engineer with **2+ years of hands-on experience** designing, developing, and deploying full-stack web applications. Built and deployed multiple complete systems - including a dealership management platform and a multi-tenant contractor application - covering authentication, role-based access control, AI-assisted features, automated reporting, and cloud deployment. Proficient in Next.js, TypeScript, Node.js, PostgreSQL, Python, and Supabase, with practical experience integrating AI using Google Gemini and Google ADK. Has delivered web projects for small business clients, gaining real experience in client communication, requirements gathering, and on-time delivery. Passionate about building well-structured, practical software and eager to contribute within a collaborative engineering team.

TECHNICAL SKILLS

Frontend: React, Next.js, TypeScript, JavaScript ES6+, HTML, CSS, Tailwind CSS, Vite, responsive dashboards, admin interfaces, role-based UI flows

Backend & APIs: Node.js, Express.js, Next.js API routes, PHP, Python, FastAPI, REST API design, middleware, validation, authentication flows, PDF/report generation

Databases & Auth: PostgreSQL, Supabase PostgreSQL, Prisma, Supabase Auth, NextAuth.js, Firebase Auth, Firestore, Firebase Storage, Oracle Database, PL/SQL basics, MongoDB, MySQL

Security & Systems: RBAC, MFA flows, role/permission checks, tenant-scoped queries, signed URLs, rate limiting, audit logs, soft delete/restore, backup workflows, password reset flows

Cloud & Tools: Vercel, Render, Netlify, Firebase Hosting, Supabase, Docker, Git, GitHub, Postman, VS Code, Figma, Draw.io, environment configuration, CI/CD awareness

AI / Automation: Python FastAPI AI services, Roboflow detection API, Google ADK, Gemini, AI-assisted business queries, n8n workflow planning, Redis, Socket.io, agentic automation research

PORTFOLIO SUMMARY

- Built and deployed business platforms for dealership operations, contractor workflows, CRM, inventory, sales, payments, reports, and admin management.
- Conducted market and workflow research through face-to-face business owner/manager discussions, SME problem analysis, Reddit problem discovery, and product planning.
- Focused on practical operational systems with auth, database relationships, dashboards, reports, permissions, AI-assisted workflows, and deployment.

PROFESSIONAL EXPERIENCE

Full Stack Engineer & Business Systems Developer - Independent, Self-Learning & Product-Based Development | Jan 2023 - Present

- Built full-stack and business-oriented software projects from self-directed learning, small web project experience, market research, and real workflow analysis.
- Designed product workflows, database models, backend API logic, protected dashboards, authentication, role-based permissions, reporting, exports, and deployment setups.
- Implemented production-style features including RBAC, MFA/security flows, audit logs, soft delete/restore, PDF generation, signed storage URLs, rate limiting, tenant-scoped data access, and admin dashboards.

MAJOR PROJECTS

Auto Prime - Dealership Management System | Live Deployed

Next.js | TypeScript | Node.js | Express | Python | FastAPI | Supabase PostgreSQL | Supabase Auth | Google ADK/Gemini | Tailwind CSS | Vercel

- Built and deployed a full dealership operating platform after face-to-face discussions with local dealership owners/staff to identify problems in stock tracking, paper-based deals, manual commissions, finance handling, and weak reporting.
- Developed a public vehicle website and secure admin console covering inventory, CRM leads, customers, deals, payments, finance, trade-ins, commissions, expenses, employees, reports, backups, trash/restore, audit logs, MFA, and permissions.
- Implemented admin, manager, salesperson, and accountant roles with granular controls for profit visibility, price editing, report exports, backups, audit logs, discount approval, customer blacklisting, and destructive actions.
- Built deal/payment workflows for reservations, partial/full payments, finance disbursement, delivery checklists, payment status updates, profit calculation, and automatic commission generation.
- Integrated a read-only Python FastAPI + Google ADK/Gemini AI assistant for dealership queries such as inventory, unpaid deals, stock aging, salesperson performance, and monthly sales summaries.

TradeTrack - Multi-Tenant Contractor SaaS | Live Deployed

Next.js 14 | TypeScript | Prisma | PostgreSQL/Supabase | NextAuth.js | Tailwind CSS | Zod | Vitest | Python FastAPI Worker | Docker | Vercel

- Built and deployed a multi-tenant SaaS product for solo contractors and service businesses after analyzing contractor workflow problems from Reddit discussions, freelancer pain points, and manual tracking patterns across notes, WhatsApp, spreadsheets, and PDFs.
- Developed the core workflow from client to quote to job to invoice to reminder to payment with tenant dashboard, clients, jobs, quotes, invoices, payments, reminders, settings, and platform admin tools.
- Implemented organization-based tenant architecture where users authenticate as people and access business data through organization membership, tenant routing, and role permissions.
- Designed PostgreSQL/Prisma models for users, organizations, memberships, clients, jobs, quotes, invoices, payments, documents, reminders, email logs, admin actions, and rate-limit counters.
- Added tenant-safety controls using organization-scoped queries, middleware slug resolution, session/membership checks, signed Supabase Storage URLs, API rate limiting, and RLS-aware transaction context.
- Built server-side PDF generation, private storage uploads, short-lived signed URLs, Vitest coverage for business workflows, and a Python FastAPI invoice worker prepared for summary/overdue processing.

MAMMOCARE - AI-Powered Mammogram Guidance Platform | Team Project

HTML | CSS | JavaScript | Firebase Auth | Firestore | Firebase Storage | Python | FastAPI | Roboflow API | Canvas API | jsPDF | html2canvas

- Contributed to a university team project for AI-assisted mammogram screening support with patient, doctor, and admin portals.
- Built role-based workflows using Firebase Authentication and Firestore, allowing patients, doctors, and admins to access separate dashboards.
- Integrated a Python FastAPI AI proxy that accepts mammogram uploads, forwards images to a hosted detection model, and returns bounding boxes, finding classes, and confidence values.
- Implemented patient report workflows including annotated mammogram rendering on HTML canvas, saved report history, Firebase Storage uploads, PNG download, and PDF report generation.
- Added doctor booking and realtime communication workflows where patients can request appointments, doctors can accept/complete bookings, and accepted sessions unlock chat with text, image, and PDF attachments.

ADDITIONAL PROJECTS & RESEARCH

Urban Food E-Commerce - Database Warehouse Module Project | Team Project

PHP | Oracle Database | PL/SQL | MongoDB | HTML | CSS | JavaScript | Dompdf

- Contributed to a university Database Management 2 module project for an online fresh-food ecommerce platform with cart, wishlist, checkout, order history, and admin workflows.
- Used Oracle Database for transactional data including users, products, carts, wishlists, orders, and admin-managed records; used MongoDB for feedback, product reviews, and contact messages.
- Implemented reporting features including admin dashboard summaries, PL/SQL procedure-based calculations, CSV/XML exports, and PDF feedback summaries.

Hexa AI Platform - Health Guidance & Doctor Consultation Research / Planned Build

Market Research | Product Planning | Google Cloud | n8n | Google ADK | Gemini | FastAPI | Node.js | PostgreSQL | Redis | Socket.io | Docker

- Researched a planned health guidance platform where users can track health data, book online doctor consultations, receive doctor-recommended nutrition guidance, and review food logs against calories, protein, and other nutrition targets.
- Planned AI-assisted modules for health profile analysis, doctor matching, consultation requests, food/nutrition analysis, recommendation summaries, realtime patient-doctor communication, reminders, and admin monitoring.
- Analyzed a possible implementation stack using Google Cloud, n8n, Google ADK, Gemini, FastAPI, Node.js, PostgreSQL, Redis, Socket.io, Docker, and secure API-based integrations.
- Current status: market research, problem analysis, and technical planning only; implementation not started.

EDUCATION & CERTIFICATIONS

Higher National Diploma in Software Engineering - National Institute of Business Management, Sri Lanka | 2024 - 2025

Diploma in Software Engineering - National Institute of Business Management, Sri Lanka | Jan 2023 - Jan 2024

Career Essentials in Generative AI - Microsoft and LinkedIn Certification | 2024

REFERENCE

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